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# Determining Early Modern Army Strength: The Case of Electoral Saxony



Lucian E. Staiano-Daniels

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## Abstract

Although armies grew larger during the seventeenth century, size estimates are often inaccurate. This article uses diverse sources to produce accurate troop counts for Electoral Saxony during the Thirty Years' War (1618–48), concluding that its army was large only briefly. It also discusses Saxony's overlooked role in this war. This research suggests that the military revolution thesis should be revised: army sizes did not necessarily stabilize at higher levels and war did not necessarily "make the modern state." Saxony went to war using traditional means of raising revenue: different states responded to their challenging environment differently.

A central element of the military revolution thesis is the development of unprecedentedly large armies during the seventeenth century. Forces on a scale not seen since the era of classical Rome supposedly stalked Europe, ballooning in size from war to war. The tremendous amounts of money they required are said to have spurred states to develop more centralized, efficient bureaucracies to raise and process resources: states became war-making machines. Both a means to enforce the extraction of resources and their greatest consumer, these hosts solidified the power of the sovereign against enemies foreign and domestic. But this discussion is more complicated than it seems.

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Lucian Staiano-Daniels is a Dan David Post-Doctoral Fellow at Tel Aviv University. He was educated at St. John's College and New York University, and received his Ph.D. at the University of California at Los Angeles. His work explores the daily lives of ordinary soldiers in seventeenth-century Central Europe. Methodologically he is interested in statistics, close reading, and unusual sources. He has written fifteen articles and presentations, and is currently revising a history of a Saxon regiment in Italy during the 1620s.

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Michael Roberts gave the military revolution thesis its original formulation in 1955. He thought that the Dutch *stadtholder* or chief magistrate Maurice of Nassau and the Swedish king Gustavus Adolphus led the way by switching from square formations to battle lines three men deep. Because these formations were smaller and because Gustavus also instituted the musket salvo and reintroduced the cavalry charge, which were both demanding tasks, these soldiers had to be better trained and better disciplined than soldiers had been in the Middle Ages. Roberts reasoned that Gustavus's confidence in the superiority of these new tactics, plus a strategic ambition that encompassed the entirety of Central Europe, led him to enlarge his army. Tactical changes led to a change in strategy which spurred a growth in the size of armies.<sup>1</sup> The concept of the military revolution quickly spread among historians of early modern western Europe. In an article (1976) and an influential book (1988), Geoffrey Parker pushed back the time period: he claimed the military revolution was driven by the new fortifications developed during the rise of gunpowder weapons in the early sixteenth century. These walls of these fortifications were lower than those of medieval fortresses, thicker, pointed to allow overlapping fields of fire, and bristling with cannon. They were difficult to take by storm, so sieges became more important, which demanded men and resources on a massive scale. According to Parker, this forced the development of the state.<sup>2</sup> John Lynn disagreed with Parker's interpretation of the cause of this process: he believed the growth in army size was a natural corollary to European population growth.<sup>3</sup> Whatever caused it, the thesis holds that the seventeenth century saw a vast increase in army size. We are told the Spanish army had a total of 300,000 combatants as of the 1630s, and the Dutch Republic had 60,000 as of 1606.<sup>4</sup> The French army supposedly numbered at least 115,000 infantry and 9,500 cavalry in 1634.<sup>5</sup> When they faced each other for the last time in 1631–32, the leading Imperial general Albrecht von Wallenstein and Gustavus Adolphus each allegedly fielded an army of 100,000 men.<sup>6</sup>

Certainly, by the turn of the eighteenth century European armies were far larger than they were 100 years previously. But precisely when and to what extent seventeenth-century armies got bigger—and how long they remained bigger—is a

1. Michael Roberts, "The Military Revolution, 1560–1660," in *The Military Revolution Debate: Readings on The Military Transformation of Early Modern Europe*, ed. Clifford J. Rogers (Boulder, Colo.: Westview Press, 1995), 13–19.

2. Geoffrey Parker, "The 'Military Revolution' 1560–1660—A Myth?," *Journal of Modern History* 48 (1976); Parker, *The Military Revolution: Military Innovation and the Rise of the West, 1500–1800* (Cambridge: Cambridge University Press, 1996).

3. John Lynn, "The trace italienne and the Growth of Armies: The French Case," *Journal of Military History* 55 (1991): 319–20.

4. M. S. Anderson, *War and Society in Europe of the Old Regime, 1618–1789* (Montreal: McGill-Queens University Press, 1988), 17.

5. Geoffrey Parker, ed., *The Thirty Years' War*, 2nd ed. (London: Routledge, 1997), 132.

6. Martin van Creveld, *Supplying War: Logistics from Wallenstein to Patton* (Cambridge: Cambridge University Press, 1977), 5.

difficult question. If we can meaningfully speak of a seventeenth-century "military revolution," it did not entail a simple rise in army size or centralization of the nascent state, but a cluster of slow and messy developments, starts, reversals, declines, and dead ends as well as progress. These developments were different for each polity.

Since Roberts's original claim was made, a great deal of research has been done on the military revolution in Europe, and the highest numbers for troop strength have repeatedly been subject to challenge. Some historians calculate troop strength according to official financial and military statements, such as patents to recruit or documents by heads of state requesting that troops be raised (*Bestellungen* in German), but these did not necessarily correspond to actual troop numbers.<sup>7</sup> It was recognized at the time that due to financial wastage and the normal corruption, a head of state needed to fund more troops than he or she ended up receiving.<sup>8</sup> The number of regiments or companies, after the troops were raised, can be problematic as a guide to army strength, since a company's paper strength often had little to do with its real numbers. But attempts to approximate a "real strength" by taking the theoretical strength or the estimates of other historians and subtracting a fixed percentage per unit, which John Lynn did, merely place the problem at arms' length: if the estimates upon which this approximation is based are themselves wrong, then the final result is as arbitrary as the starting point.<sup>9</sup> Attempts to find real strengths through official sources not only produce numbers that are too big, but also produce numbers that are too stable. The underlying assumption that there is "an army size" to begin with, a single number which persists over time and which contemporary observers expected to persist over time, is flawed.<sup>10</sup> As David Parrott writes, this assumption owes more to nineteenth- or twentieth-century warfare than the reality of the seventeenth century, in which "[a]rmies were institutions whose size and composition fluctuated continually."<sup>11</sup> Troop strength seesawed wildly, with frequent massive losses due to illness or desertion offset by constant recruitment. More than one campaign ended in disaster when a general's army crumbled out from under him.

Historians now agree that changes in European states, including increases in the size of armies, had no goal, and were characterized by ups and downs, stops and starts. They also agree that unit size during the seventeenth century was usually

7. John Lynn points this out for France in "Recalculating French Army Growth during the Grand Siècle, 1610–1715," *French Historical Studies* 18 (1994): 881–906.

8. David Parrott, *Richelieu's Army: War, Government, and Society in France, 1624–1642* (Cambridge: Cambridge University Press, 2006), 168.

9. John Lynn, *Giant of the Grand Siècle: The French Army, 1610–1715* (Cambridge: Cambridge University Press, 2006), 50–51. Lynn is criticized in Parrott, *Richelieu's Army*, chap. 2, 169.

10. Jürgen Pohl, "Die Profiantirung der Keyserlichen Armaden Ahnbelangendt," *Studien zur Versorgung der Kaiserlichen Armee 1634/1635* (Horn/Vienna: F. Berger & Söhne, 1994), 25–32; Cordula Kapsner, *Die bayerische Kriegsorganisation in der zweiten Hälfte des dreissigjährigen Krieges 1635–1648/49* (Münster: Aschendorff Verlag, 1997), 56–58.

11. Parrott, *Richelieu's Army*, 178.

far lower than paper strengths. But without concrete examples, this conversation remains theoretical. The only way to get reliable numbers for seventeenth-century armies is to look at lower-level sources that were produced by people in actual contact with soldiers, or that drew on data provided by people who were, such as muster rolls, payrolls, or strength returns. Several studies use sources produced by armies themselves. For instance, Jürgen Pohl draws on army lists for his figures for Imperial army strength in 1634–35, and Cordula Kapser cites army-produced sources for her figures for Bavarian army strength in the second half of the Thirty Years' War.<sup>12</sup> Bernhard Kroener's sources for the French army from 1635 to 1661 are musters, general reviews of the troops, and provisioning records.<sup>13</sup> Gregory Hanlon uses muster rolls for his study of the army of the Italian duchy of Parma in 1635 and 1636.<sup>14</sup> This article stands within the tradition of these more cautious studies. Utilizing data drawn from sources that are close to the daily life of armies, it adds to these precedents by providing troop counts which are as accurate as possible for Electoral Saxony during the Thirty Years' War.

The Thirty Years' War (1618–48) was proportionally one of the most destructive conflicts in European history. Although it was fought primarily in Central Europe, this vast and sprawling conflict involved political entities from England to what is now Russia. The war began when Protestant Bohemian leaders objected to the selection of the fervent Catholic Ferdinand of Styria (the future Ferdinand II) as heir to the Holy Roman Empire, but hostilities spread as other powers intervened. Roughly, the protagonists can be divided into Imperialist and anti-Imperialist factions: the Emperor, Spain, Bavaria, and minor Catholic powers on one side; and the rebellious Bohemian Estates, anti-Imperialist German powers, Denmark, Sweden, and France on the other. When war broke out, Saxony joined the Imperial side. Despite his Lutheran faith, Elector Johann Georg I was drawn to the Emperor as a reaction to the political and religious threat posed by the sudden rise of Calvinism, which not only was heretical from his point of view but also threatened to destabilize the status quo within the Holy Roman Empire. In addition to maintaining his hold on his territory through loyalty to the Emperor, the Elector also sought to gain land at the expense of the Bohemian rebels if possible. From 1618 until 1625, Saxon forces fought in Bohemia and in Upper and Lower Lusatia, two territories of the Bohemian Crown Lands bordering Saxony. Their efforts were successful. The Lusatias were taken over by the Hapsburg monarchy, but held by Saxony pending the Emperor's repayment of some of the Electorate's war debts. From 1625 until 1631 Saxony attempted to steer a middle course between crusading Catholicism on the one side and militaristic Protestantism on

12. Pohl, *Profiantirung der Keyserlichen Armaden*, 29–32; Kapser, *Bayerische Kriegsorganisation*, 218–49.

13. Bernhard Kroener, "Die Entwicklung der Truppenstärken in den französischen Armeen zwischen 1635 und 1661," in *Forschungen und Quellen zur Geschichte des Dreißigjährigen Krieges*, by K. Repgen (Münster: Aschendorff Verlag, 1981).

14. Gregory Hanlon, *The Hero of Italy: Odoardo Farnese, Duke of Parma, His Soldiers, and His Subjects in the Thirty Years' War* (Oxford: Oxford University Press, 2014), chap. 2.

the other, remaining neutral and pushing for peace in the Empire. This failed; in 1631 the Electorate reentered the fight as a reluctant Swedish ally. In 1635, Saxony rejoined the Imperial side and the Emperor ceded the two Lusatias to it. Saxony remained an Imperial ally until the end of the war.

Electoral Saxony was the most powerful single Protestant state in the Holy Roman Empire, a mid-level German power comparable to large and wealthy Bavaria in strength and preeminence. It is also well documented: the holdings of the Saxon Hauptstaatsarchiv in Dresden remained almost entirely intact during World War 2, making it an unusually rich source. Yet before World War 2, historians viewed Saxony during the Thirty Years' War as militarily ineffective and its foreign policy as pointless.<sup>15</sup> During the Cold War, few historians paid much attention to it at all, for it was difficult for westerners to study Saxony and historians in the German Democratic Republic were not encouraged to investigate seventeenth-century military history. However, Saxony complicates narratives of seventeenth-century military progress since it is an outlier to the European trend of military growth and state consolidation. Historians argue that the demands of larger armies supposedly led to the growth of the modern bureaucratic state, since governments raised new taxes to support them and employed ever-greater numbers of officials to direct the movement of the resources they raised. Since this machinery of state was subordinate to the sovereign rather than the Estates, state power also supposedly grew more centralized.<sup>16</sup> Yet instead of a rise in army size entailing the modernization of the state, what happened in Saxony in the seventeenth century was an increase and then decrease in army size accompanied by the retention of existing practices of administration and resource extraction. War did not "make the state" in Saxony: instead, it led to unprecedented strain on the Electorate and even made financial administration seem impossible to authorities in some territories. But despite taking a beating, Saxony managed to weather the Thirty Years' War intact. More than that, this article argues that the Saxon army remained a force to be reckoned with throughout the war.

Examining the timing and rate of changes in the size of the Saxon army also allows us to shed new light on the effects of important battles of the Thirty Years' War like Breitenfeld (1631) and Wittstock (1636), or events like the Pomeranian campaign of the controversial Imperialist general Matthias Gallas (1637–38). It also reintroduces contingency and individual decision-making into this discussion, which has hitherto focused on structural explanations for historical change. While this article does not discount the results of broad developments such as supply problems or lack of funding, it also touches on the results of the Elector's decisions to retain or demobilize forces, or ill-luck like a defeat in battle or a ruinous campaign.

The data presented in this article are drawn from a wide variety of source material. The largest source of troop counts is a pair of volumes that were probably

15. C. V. Wedgwood, *The Thirty Years' War* (New York: New York Review Books Classics, 2005), 62–65.

16. Charles Tilly, *Coercion, Capital, and European States, AD 990–1992* (Oxford: Blackwell, 1992).

created on behalf of the Electoral administration in the 1650s, *Churf. Durchl. Zu Sachsen etc Erste und Andere Kriegsverfassung Nach entstandener Unruhe im Königreich Böhmen*<sup>17</sup> and *Kriegs Verfassung de Anno 1618 usq Anni 1650 vol II*.<sup>18</sup> The first volume contains a list of the mercenary regiments raised in Saxony during the Thirty Years' War, and the second volume contains strength reports for all Saxon mercenary troops from the 1630s to the end of the war. The latter is a collection from other documents which probably no longer survive, like muster rolls, before-battle or after-battle reports, lists made when "coming out of quarters" at the beginning of the fighting season, and letters from colonels enumerating their forces. These volumes may have been put together by the same anonymous compiler. Although no explicit information survives about this person, he may have been a dragoon or somehow connected to a dragoon squadron, since he wrote more about dragoons than other branches of service. The first volume contains brief, impersonal statements of fact for infantry and cavalry, but longer entries, judgments about individual officers' performances, and colorful personal anecdotes for dragoons. For instance, Constantine Lehman, captain of a squadron of artillery dragoons, commanded this unit from its founding in October 1634 until December 1649, when he had his girlfriend's husband shot in the street. On 1 March 1650, he, she, and the triggerman were executed in Dresden.<sup>19</sup>

This article also draws on 112 company muster rolls and three payrolls from the 1620s as well as 34 company rolls from the 1640s: a substantial fraction of the 173 total Thirty Years' War mercenary rolls remaining in the Saxon State Archives. In addition, this study uses strength tables from the 1640s, published letters, and published troop numbers from the work of Roland Sennewald.<sup>20</sup> (These sources often overlap, giving us the opportunity to check them against one another.) Seventeenth-century muster rolls are useful not merely for the data they contain but also because they are often first-hand accounts in their own right: company officials who were both diligent and untidy kept notes in the margins of their own rolls, covering the comings and goings of the soldiers, who is sick, who has died, who needs new equipment, which cavalymen are unmounted, and so on. Some historians claim that because soldiers came and went continuously, seventeenth-century military authorities did not record the death or absence of individual common soldiers.<sup>21</sup> But companies required accurate knowledge of how many soldiers there were, how healthy they were, and how well equipped they

17. Sächsisches Hauptstaatsarchiv Dresden (SHStADr) 11237 10831/1 *Churf. Durchl. Zu Sachsen etc Erste und Andere Kriegsverfassung Nach entstandener Unruhe im Königreich Böhmen*.

18. SHStADr 11237 10831/2 *Kriegs Verfassung de Anno 1618 usq Anni 1650 vol II*.

19. "Da er kurz vorher seiner Concubinen Ehemann uff der strasse erschossen lassen undt folgens nebenst derselben undt dem einen thäter den 1 Marty 1650 uffn neumarck alhier justificiret wurde." SHStADr 11237 10831/1.

20. Roland Sennewald, *Das Kursächsische Heer im Dreissigjährigen Krieg 1618–1648* (Berlin: Zeughaus Verlag, 2013).

21. Gregory Hanlon, *Italy 1636: Cemetery of Armies* (Oxford: Oxford University Press, 2016), 176.

were. This was necessary to house and feed soldiers and their families and to field combat-ready forces. It is more likely that these historians' lack of data reflects the ephemeral nature of the records. Muster rolls were working documents, sometimes written over multiple times. Some housing records were maps of villages sketched on little slips of paper.<sup>22</sup> The records that we can see and analyze are probably only a tiny fraction of a vast accumulation of ad hoc paperwork that has not survived, the quotidian chatter of companies constantly monitoring themselves.

What is essential here is not that some sources are primary and some are secondary; it is who made the primary sources we look at and what those people were able to know. Although no primary document is infallible, documents produced by low-ranking people who were in daily contact with armies probably provide the most accurate troop numbers it is possible to obtain. Sources like rolls, provisioning records, death lists, or accounts of shortfalls of regiments give troop numbers that are lower and less stable than the numbers in higher-level government sources like contracts, *Bestallungen*, letters by heads of state, or paper strengths. Although the latter are also primary sources, they depict what those who hired mercenaries wanted, not necessarily what they got—or could keep.

A review of some early seventeenth-century military terms will help the reader compare paper strength to real strength. A company was the smallest administrative unit. On paper, an infantry company comprised 314 combatants in the 1620s and 214 in the 1630s and 1640s. The human beings serving in cavalry companies are more difficult to count since some cavalry records tracked only the number of horses: the 100 horses which made up a cavalry company on paper probably corresponded to around 30 to 50 fighting men in addition to servants and “boys.” A regiment was an administrative unit comprising more than one company, often around ten, so its paper strength was often around 3,000 in the 1620s and around 2,000 in the 1630s and 1640s. A free company is a company which was not part of a regiment and which was led by a colonel. Many dragoons were organized in squadrons, a word which was often used for a company of mounted troops. Some dragoon regiments were led by captains. In the 1620s, a Saxon infantry company was composed on paper of eighty pikemen, twenty halberdiers, and 200 musketeers; infantry in the 1630s and 1640s were composed of two-thirds musketeers and one-third pikemen. Saxony never fielded lancers during this war: its cavalry were arquebusiers, lightly armored or unarmored troopers armed with a short-barreled longarm; and cuirassiers, troopers armed with a pair of pistols and encased in plate armor from head to knees—when they were kitted out at all, for armor and pistols were expensive and could be damaged. Two cuirassier rolls from the 1630s have “nicht montiert” (unequipped) written beside almost every name.<sup>23</sup> Dragoons were mounted musketeers, who travelled and foraged on horseback but dismounted to fight. They were more flexible than cavalry and faster

22. *RELATION Chur Fürstl Durch in Sachsen Reise mit die Armee vom 10 Decbr. 1634 bit 7 Januar 1635*, SHStADr 11237 10831/3.

23. SHStADr 11237 10841/2-2 and 11237 10841/2-3.

than infantry, but neither of these branches had much respect for them: when the dragoon squadron of the talented Andreas Masslener, nicknamed “Ungar,” did exceptionally well, it was reequipped as cavalry.<sup>24</sup>

The findings that inform this article are compiled and simplified in four tables, listed in the appendix. When we add up the figures from these tables, we obtain an approximation of total Saxon army strength during the Thirty Years’ War, which is tracked in the following graph (see facing page).

This is a line graph representing the total strength of Saxon cavalry, infantry, and dragoons month by month from 1618 until 1651, with a break from 1625 to 1631 when Saxony had no army. Since cavalry tracks the number of horses while infantry and dragoons track the number of soldiers, the three branches are charted separately.

From 1618 to 1625, real company strength came close to paper strength. It was also stable; absences were frequent, but they were readily replaced. Although many soldiers stayed around for years, whether in the army in general or in the same unit, a constant turnover can be seen in the rolls from this period as some soldiers left or died and others entered. While it was a heavy financial burden for Saxony, the 1618–25 conflict in the Bohemian territories was more limited than the conflict that was to come after 1631. According to muster rolls, most Saxon forces were also paid at least four times a year during the 1620s. However, the sudden drop in army size in 1623 militates against the contention that one element of the military revolution was the creation of early standing armies: instead of maintaining its forces as a proto-standing army, Saxony hurried to demobilize these expensive drains on its economy as soon as possible when hostilities in the region came to an end, and then raised new troops only when they were needed again. Similarly, after 1625 the Saxon army was disbanded almost totally; recruitment began again in 1631.

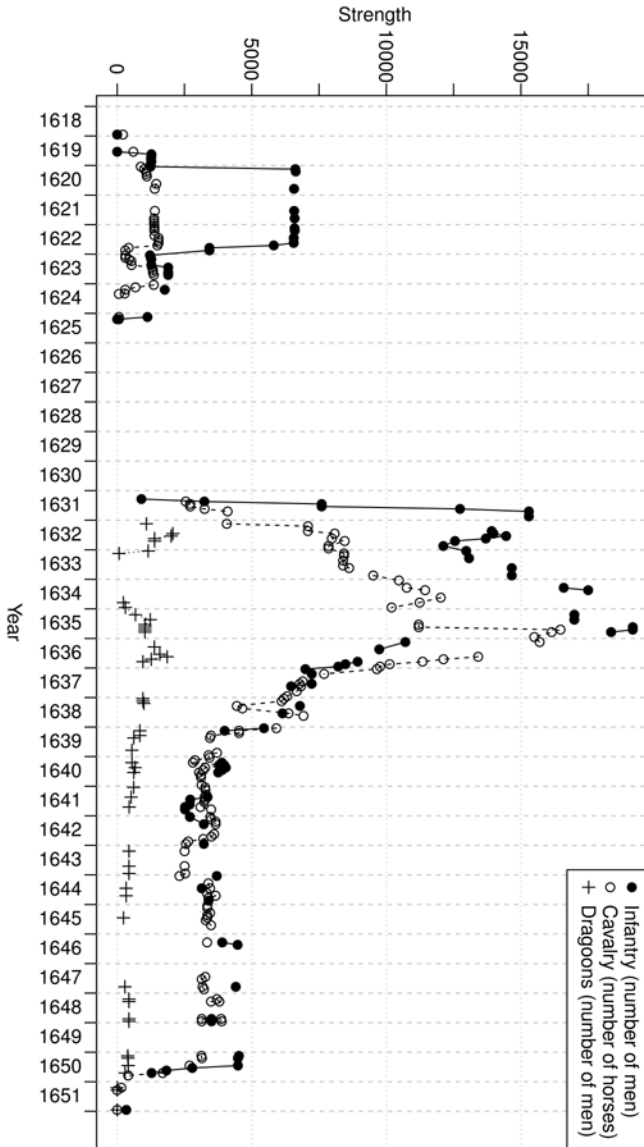
The increase in army size through the first half of the 1630s and its peak in 1635 are in line with what we already know about the Thirty Years’ War. So are the smaller but more numerically stable forces of the 1640s, characterized by a comparatively high proportion of cavalry and dragoons.<sup>25</sup> Regimental strength findings from 1631 to 1651 corroborate the findings of other historians, as well as anecdotal evidence from first-hand accounts of early modern armies: in this period, troop strength fluctuated constantly, and was almost always far below the paper strength. But while unit strength changed frequently, the seasonal fluctuations that John Lynn has remarked on (in which a force recruits over the winter break, is strongest when coming out of quarters in the spring, and then declines all summer and fall) are not always apparent in the tables.<sup>26</sup> Regiments sometimes declined during the fighting season, but they sometimes got larger as well, and sometimes their strengths rose and fell. Once a regiment had been raised recruitment appears

24. SHStADr 11237 10831/1.

25. Peter H. Wilson, *Europe’s Tragedy: A New History of the Thirty Years War* (London: Penguin UK, 2010), 623.

26. John Lynn, *The Wars of Louis XIV: 1667–1744* (London: Routledge, 2014), 369.

**Figure 1:** Running total of Saxon army strength by branch of service, values from Tables 1–4.



to have been constant, which is attested by the marginal notes in a muster roll for Ernst von Günterrode's free company from 1619 to 1620 that several soldiers are absent because they are "vff der werbung" (gone recruiting).<sup>27</sup>

27. SHStAdr 11237 10839/6.

Officers paid attention only rarely to the theoretical strength of regiments; instead, a regiment's strength was compared to its past strength at the most recent previous mustering. If a regiment was "found complete" it meant that both figures matched, unlike August Adolph von Drandorf's infantry regiment on 27 April 1646, which was "supposed to be 860," but "what was found was 841."<sup>28</sup> (This was in a decade when the paper strength of an infantry regiment was still 2,000.) As long as we are describing the daily practice of this army, it may be inappropriate to speak of a "standard" unit size at all. Considering the number of strength reports and their centralized location, information like that collected in *Kriegs Verfassung de Anno 1618 usq Anni 1650* may have been monitored at the time by commanders while planning expeditions, without having to rely on paper strength estimates.

The precise timing of changes in army size also reveals some interesting insights. For instance, the first battle of Breitenfeld does not appear to have been very consequential. This battle took place on 7 September (old style) 1631 near Leipzig between an Imperialist army under the Count of Tilly and a joint Swedish/Saxon army. Although the Swedes eventually won, the newly enlisted Saxon troops collapsed early in the fight and ran.<sup>29</sup> Breitenfeld has long had a central place as a symbol of Gustavus Adolphus's dynamic leadership; both the Saxons and the Imperialists appear in the usual narrative as a foil for the ascendancy of Swedish power. But although poor performance here cast a shadow over our view of Saxony during the entire war, it did not hit Saxony where it hurt, which was troop strength and the ability to raise and maintain an army. Men who run can also come back, and although Saxon casualties were 3,000, regiments continued to be raised month after month.<sup>30</sup> By June 1632, the Saxon army was about 24,000 strong.<sup>31</sup> Breitenfeld does not appear to have made much of an impression on the anonymous dragoon either; the only time he says anything definite about it is when he specifies that the First Life Company was not part of the Electoral Life Regiment of Horse:

this company was not counted among the first five Taubish companies [those belonging to the Electoral Life Regiment of Horse, also known as Colonel von Taube's regiment], it made up the sixth, and at the first battle before Leipzig they marched out with the Taubishes and fought, and there Rittmeister Marschall was shot by a cannon and therefore Lieutenant Georg Herrfurth became Rittmeister. So it is that the company did not remain

28. Infantry regiment number 2 in Table 3; it began its career as Schwalbach's Regiment. SHStADr 11237 10841/13.

29. Thomas M. Barker, *The Military Intellectual and Battle: Raimondo Montecucoli and the Thirty Years War* (Albany: SUNY Press, 1975), 174–81; Parker, *The Military Revolution*, 23.

30. Barker, *Raimondo Montecucoli*, 181; Wilson, *Europe's Tragedy*, 474.

31. Wilson gives this figure as well, but he says it also included garrison troops, which it may not have. *Europe's Tragedy*, 477.

with the Taubishes, but they were apart and His Elector's Life Company and the entire time they were at the Hoffstadt under the Hoffmarschall's command.<sup>32</sup>

The honorable deeds of this company—and making sure they are attributed to the correct group of people—sound like concerns of more pressing importance to this chronicler than the course of the battle at which they were performed.

Unlike Breitenfeld, the battle of Wittstock in 1636 and the Pomeranian campaign of 1637–38 crippled the Saxon army. Saxony rejoined the Imperial side in 1635. The agreement they signed with the Emperor, called the Peace of Prague, stipulated that while Imperialist forces and Bavaria defended the Rhine and aided Spain in the west of Germany, Saxony was to deal with Sweden, whether by negotiation or force. But when the Elector of Saxony offered Sweden financial compensation to withdraw, the Swedes rejected the deal.<sup>33</sup> By September 1636 the Saxons were looking for a fight, and Wittstock was a battle they had to win. As in many Saxon military endeavors, they failed.

The battle of Wittstock took place on 24 September (old style) 1636, between an anti-Imperialist force under Johan Banér and Alexander Leslie, and an Imperialist/Saxon force under Melchior von Hatzfeld. Although the battle was hard fought, it too ended in a Saxon defeat.<sup>34</sup> Yet whatever their performance five years earlier, this time the Saxons held up under pressure; Banér, who led his cavalry in person on the traditional right flank, wrote later that “there was not one of his squadrons that did not have to engage them at least six times, or as many as ten.”<sup>35</sup> There has been little consensus about the number of troops on the field, which may be impossible to determine.<sup>36</sup> However, although we cannot say with certainty how many Saxon cavalry troopers fought at Wittstock, it is certain that between the fifth and the twenty-first of September (old style), 11,365 Saxon cavalry horses were counted. The exact relationship between the number of horses and the number of riders was not fixed: this number of horses may have corresponded to as many as 10,000 combatants. Although many of the hungry Saxon troops left to forage right before the battle—accounts mention foragers moving together in bands of up to 300 men—it is possible that half the

32. SHStADr 11237 10831/1, 48r–48v.

33. Peter H. Wilson, “Meaningless Conflict? The Character of the Thirty Years War,” in *The Projection and Limitations of Imperial Powers, 1618–1850*, ed. Frederick Schneid (Leiden: Brill, 2012), 28.

34. Wilson, *Europe's Tragedy*, 580–83.

35. Steve Murdoch, Kathrin Zickermann, and Adam Marks, “The Battle of Wittstock 1636: Conflicting Reports on a Swedish Victory in Germany,” *Northern Studies* 43 (2012): 80.

36. Rudolf Schmidt, *Die Schlacht bei Wittstock. Ein Beitrag zur Geschichte des dreißigjährigen Krieges* (Halle: Gesenius, 1876), 60–63; Hans Delbrück, *The Dawn of Modern Warfare: History of the Art of War*, vol. 4 (Lincoln: University of Nebraska Press, 1990), 213; William P. Guthrie, *The Later Thirty Years War: From the Battle of Wittstock to the Treaty of Westphalia* (Westport, Conn.: Greenwood Publishing Group, 2003), 62–64, 42; Barker, *Raimondo Montecuccoli*, 206–12.

combined force's cavalry was Saxon.<sup>37</sup> The infantry and dragoons at Wittstock were not enumerated with such specificity in the surviving documents; the closest available numbers for most infantry regiments are from February or May 1636. The only unit here with a sure number from September is Ungar's Regiment of dragoons, since "Ungar" Masslener wrote that he had 309 dragoons in a letter dated 12 September.<sup>38</sup> Saxon infantry probably numbered about 6,000, with a smattering of dragoons.

Certainly it was a savage fight, possibly the hardest cavalry fighting of the war: Banér wrote that when the Imperialists and Saxons regrouped to react to his attack, "immediately a cruel fight sprang up, the like I had not witnessed during the time of my whole life."<sup>39</sup> Joint Imperialist/Saxon casualties were about 3,000–3,500 and about 2,000 prisoners, but according to Sennewald many prisoners escaped and made it back to their own companies. The Saxons probably accounted for about half the losses.<sup>40</sup> Although the retreat almost certainly did not cost as many of his enemies' lives as Banér claimed, it was deadly too: flags and baggage were lost, the soldiers scarcely had anything to dress themselves in, the infantry were sent on boats over the Elbe, and the cavalry swam the Havel and headed towards Magdeburg. Many cavalry regiments had only 40 horses, and there were as few as three or four troopers in many companies. Seven regiments had only two pack wagons among them.<sup>41</sup> The usual number is one per company. By 19 October the Rochau regiment had fallen from 856 horses to 580; by November, Baudissin's regiment was down from 855 horses to 406. The number of horses tracks the invisible number of cavalymen, which must have fallen without respite for most regiments until spring: the Rochau regiment had 470 by the eighth of November and 280 at some later point without a date, but that was written down in quarters that winter or fall.

But what really traumatized the anonymous dragoon was the Pomeranian campaign that followed. During 1637 and 1638, Matthias Gallas, then the Imperialist commander-in-chief, led an Imperial/Saxon task force north through Mecklenburg, Pomerania, and Brandenburg against Banér. At first Gallas succeeded in pushing his opponent back to the Swedish forts on the sea coast, many of which he took, but he could not secure winter quarters or supplies from the powerful cities in the region. Although a recent biography argues that this tragic figure does not deserve his later reputation as a bungler, his army disintegrated in Pomerania, which had been laid to

37. Sennewald says that the archival sources demonstrate a Saxon cavalry and dragoon total of more than 11,500 soldiers before the foragers left. *Kursächsische Heer*, 275.

38. Sennewald, *Kursächsische Heer*, 280.

39. "... auch alsobalt zu einem crudelen fechten gediegen, das ich die zeit meines lebens dergleichen nicht beygewohnt," Sennewald, *Kursächsische Heer*, 290.

40. Sennewald, *Kursächsische Heer*, 293.

41. Letter of a Saxon cavalry colonel to Colonel Kalckstein, 3 October (old style) 1636. Schmidt, *Die Schlacht bei Wittstock*, 81–82. Banér claimed that his enemy suffered 5,000 lost in the retreat, which Guthrie disputes. *The Later Thirty Years War*, 57.

waste.<sup>42</sup> The dragoon was still struck by the Pomeranian campaign years later: "In the yr. 1637 von Wolframsdorf's [cavalry] regiment marched with the Impl. Main Task Force [Armada] into Pomerania, and when things were done over there it had been completely driven down to the earth."<sup>43</sup> This regiment had 38 horses in it at the end of 1637.<sup>44</sup> The Mitzlaff regiment of infantry, which had joined the Saxon army back in August 1635 when its colonel defected from the service of Saxe-Weimar, came back from Pomerania in April 1638 with 56 soldiers in it, 10 of whom were officers. In March of the previous year, this unit numbered 285.<sup>45</sup> Officers were the least likely to desert because they were more closely connected to their superiors through personal ties; they also had more to lose and more chances of actually making money from military service. All but the recorded 10 of this regiment's officers may have been dead by April 1638. When many of the other regiments marched out of Pomerania at the end of the year, Dietrich von Starschedel's regiment of infantry remained until January 1639 at the city of "Lois [?]" to besiege it "and were ruined by it"; they came back 30 strong. Their last previously recorded strength, in March 1637, was 586.<sup>46</sup> Everyone in the Rochau regiment received last rites in July 1638. Maybe they knew what was coming. They got off lightly: when they "came back up" from Pomerania and mustered in again at Mühlberg the next February, they were down to 291 horses from 653.<sup>47</sup>

This time Saxon troop strength did not recover: ten cavalry and eight infantry regiments were reduced or reformed in 1638 and 1639, and troop numbers continued to decline until early 1640. Not only did Wittstock and the Pomeranian campaign cost more Saxon lives than Breitenfeld, but there are more factors at work here than the results of battles and campaigns. The number of troops began to decline a year before the battle of Wittstock: the Saxon army was the largest it ever got during the war around September 1635—16,471 cavalry horses, 19,156 infantry, and 1,031 dragoons.<sup>48</sup> This army took the field in August 1635,

42. Robert Rebitch, *Matthias Gallas (1588–1647), Generalleutnant des Kaisers zur Zeit des Dreissigjährigen Krieges: Eine militärische Biographie* (Münster: Aschendorff Verlag, 2006), 170–85.

43. "Ao 1637 marchiret mit der Keyserl haupt Armade das Wolframsdorffische Regiment in Pommern, undt weil es allda vollendet gar zu Boden geführt wardt." SHStADr 11237 10831/1, 90 r.

44. Sennewald, *Kursächsische Heer*, 562. The regiment was founded as the Graf Solms Regiment; number 7 in Table 2.

45. SHStADr 11237 10831/1, 33 v.

46. SHStADr 11237 10831/1, 33 r.

47. After listing the number for 11 July, the chronicler writes that the regiment "bekahm aber derer endes hernachmals die Letzte Ölung, undt alß es wieder herauff kahn undt Ao. 1639 bey Mühlberg gemustert den 12 Februarii alda nicht mehr dem 291 Pf." SHStADr 11237 10831/2, 89.

48. This is up slightly from the numbers Lothar Höbelt cites from an Imperialist source enumerating the Saxon army, dated 27 July of the same year: 16,426 cavalry, 19,290 infantry, and 758 dragoons. In two months, the Saxon army had collected 273 dragoons. *Von Nördlingen bis Janckau: Kaiserliche Strategie und Kriegführung 1634–1645* (Vienna: Heeresgeschichtliches Museum, 2016), 66 fn. 148.

beginning a series of operations which lasted until late December and began again the following spring. They were badly led and poorly supplied, and unsuccessful.<sup>49</sup> Yet military setbacks alone do not account for the plunge in Saxon army strength. We should also briefly discuss Saxon finances and administration during the war.

Early modern states varied in the extent to which they grew more centralized or brought military matters under state control. Saxon Electoral authorities appear to have been deeply involved in war making from the beginning of the Thirty Years' War to its end. Primary sources from the 1620s show that an Electoral commissioner monitored the pay of mercenary units, signing off on many musters. Many Saxon officers belonged to families that served the Electoral administration, such as the von Pölnitzes, or both the Electoral and Imperial administrations, like Wolf von Mansfeld. Members of the ruling Wettin dynasty commanded some companies. On the other hand, during the war the Saxon army was not placed under permanent Wettin control, in contrast to the placement of the Imperialist army under Hapsburg control. Moreover, in contrast to what happened in some other German political entities like Brandenburg, the power of the Saxon Estates compared to the Elector expanded after the war.<sup>50</sup> Governmental administration also remained decentralized, which is why it was easy to partition Electoral Saxony after Johann Georg I's death in 1656. In sum, as Saxony responded to the Thirty Years' War, an existing system was repurposed to face new challenges. This was not always successful.

Saxony went to war by raising money with traditional means: the Elector calling the Estates to institute taxes on an ad hoc basis or extend existing taxes, and going into debt. From 1631 to 1635 the Leipzig Colloquy (a group of moderate Protestant powers which tried to remain neutral, led by Saxony) had also been responsible for raising money to support the Saxon army. These measures did not raise enough.<sup>51</sup> Since scarcely any money came from the Elector, Saxon colonels funded themselves with the typical patterns of military enterprise that characterized this war: paying to raise their own troops up front, then extorting contributions out of the areas in their path to cover the cost.<sup>52</sup>

Although the armies raised in 1618–20 and 1623 were small compared to the troops raised later, debts had been piling up since the early 1620s. The exact total of Saxon debt is obscured by the fact that finances were handled by several administrative organizations at once. Most tax income was handled by the Saxon Kammer, the privy council, but the debt held by the Elector himself was managed by a specially designated Obersteuerkollegium, a supervisory committee which taxed the trade at the Leipzig fair but did not have the authority to execute orders. This debt was supposed to be covered by ad hoc taxes, like the one on beer and wine.

49. Sennewald, *Kursächsische Heer*, 249–55.

50. Fritz Kapahan, "Kurfürst und kursächsische Stände im 17. und beginnende 18. Jahrhundert," *Neue Archiv für sächsische Geschichte* 43 (1922): 62–79, 69–71.

51. Sennewald, *Kursächsische Heer*, 595.

52. Sennewald, *Kursächsische Heer*, 582.

In 1620, the debts reported by the Obersteuerkollegium alone were 5.4 million gulden.<sup>53</sup> (Most regular pikemen made about 8 gulden a month, and military authorities estimated it took 6 gulden to feed and house a soldier for a month. Oats for one horse cost slightly more than 4.5 gulden per month.) By 1628 the financial situation had dramatically worsened: the Obersteuerkollegium's debts were about 6.2 million gulden but the Kammer's debts had reached 7 million gulden, which means that the Electoral Saxon economy was encumbered with debts worth over 13 million gulden in total.<sup>54</sup> By 1646, total debts had ballooned to about 22 million gulden, which had gone almost entirely unserviced.<sup>55</sup> The difficulty of figuring out the state of Saxon finances during the war is compounded by inflation, which hit Saxony especially hard during the so-called Kipper- and Wipperzeit, an economic crisis in the early-to-mid 1620s when currency became debased throughout the Holy Roman Empire. At some point in the late 1640s—one historian writes that it was pointless to try to determine exactly when—Saxony went bankrupt.<sup>56</sup>

The damages of the war compounded these financial problems. Armies lived, travelled, and fought in Saxon territory from late 1631 until the end of the war: this not only was a burden in itself, but also made it more difficult for the Estates to attempt to pay off debt by extending or increasing taxes on commodities, since by 1633 the villages in many places were only partially populated.<sup>57</sup> The Peace of Prague brought no relief, since enemy and Imperial troops still occupied the land as well as Saxon soldiers. Moreover, this agreement established a system whereby the territorial powers of the Empire would fund the Imperial army, which also cost Saxony money.<sup>58</sup> Saxony probably reached its economic low point in 1643–44: a Swedish garrison had been quartered in Leipzig since autumn 1642 and launched repeated raids from that city, making large parts of the Electorate financially ungovernable.<sup>59</sup> These years saw the lowest count for Saxon cavalry, the branch of service with the most complete records, until demobilization began in the early 1650s. Ironically, the punishing treaty of Kötzschenbroda, signed in September 1645 between Saxony and Sweden, coincided with a steady level of Saxon cavalry strength and preceded a small increase in infantry strength. Even though the treaty forbade Saxony from raising any new regiments and demanded 11,000 thalers a month for the upkeep of the Swedish garrison in Leipzig (a thaler is 1.5 gulden), this armistice also stopped that garrison's raiding, which gave Saxony some breathing room.

53. Uwe Schirmer, *Kursächsische Staatsfinanzen (1456–1656): Strukturen—Verfassung—Functionseliten* (Stuttgart: Franz Steiner Verlag, 2006), 805.

54. Schirmer, *Kursächsische Staatsfinanzen*, 806.

55. Schirmer, *Kursächsische Staatsfinanzen*, 817.

56. Schirmer, *Kursächsische Staatsfinanzen*, 817.

57. Schirmer, *Kursächsische Staatsfinanzen*, 810.

58. Hubert Salm, *Armeefinanzierung im Dreißigjährigen Krieg: Die Niederrheinisch-Westfälische Reichskreis 1635–1650* (Münster: Aschendorff Verlag, 1990), 164–76.

59. Schirmer, *Kursächsische Staatsfinanzen*, 816–17.

But by the early 1640s, the Electorate could no longer support the troops through state revenue and Saxon soldiers subsisted directly off their surroundings. Nineteenth-century military theorists as well as historians down to at least the 1970s interpreted seventeenth-century strategy as bankrupt because it was supposedly overly dominated by logistics. This is an anachronistic reading of the seventeenth-century military situation through a nineteenth- or twentieth-century lens.<sup>60</sup> Supply was more important to seventeenth-century military operations than any other factor, but it makes little sense to say that it was “overly” important: early seventeenth-century Europe was comparatively sparsely populated, with poorly developed agriculture, and it was difficult to support armies in the field. The most pressing challenge for any seventeenth-century commander was keeping the troops fed and the army together. (While Electoral Saxony tried to institute a regional system of military magazines after 1635, this remained rudimentary and may not have been used.<sup>61</sup>) Commanders did what they could within the physical constraints on what they could achieve, including the size of the forces they could raise. In order to live off the land Saxon soldiers needed to engage in a multitude of complex and ongoing interactions with the environment through which they moved, such as stealing or extorting food, reaping and milling grain and baking bread, finding and carrying water, chopping and hauling vast quantities of firewood, and so on. For instance, considering the numbers of mounts for officers and the cavalry as well as draft animals for artillery and baggage, common soldiers probably spent much of their time gathering, transporting, and storing fodder, or fighting off enemies who were attempting to take advantage of these scattered foraging expeditions.<sup>62</sup> Raiding and counter-raiding grew in importance during the later years of the war; notes in some muster rolls reveal this, as they record which soldiers have been “commanded to” some place or another separately from the main group.<sup>63</sup> The size of an early modern army did not just affect its ability to meet the enemy or work with allies, but also affected its ability to move through a territory and

60. On the supposed bankruptcy of seventeenth-century strategy, see Clausewitz, *On War*, ed. and trans. M. Howard and P. Paret (Princeton, N.J.: Princeton University Press, 1976), 330, 553; Delbrück, *The Dawn of Modern Warfare*, 293–315; Michael Roberts, “Gustav Adolf and the Art of War,” in *Essays in Swedish History* (London: Weidenfeld & Nicolson, 1967), 72–75; Parker, “The ‘Military Revolution,’ 1560–1660: A Myth?”; and van Creveld, *Supplying War*, 17. Historians who have emphasized the importance of logistics in the seventeenth century include Geza Perjés (“Army Provisioning, Logistics and Strategy in the Second Half of the 17th Century,” *Acta Historica Academiae Scientiarum Hungaricae* 16 [1970]); Bernhard Kroener (*Les Routes et les Étapes: Die Versorgung der französischen Armeen in Nordostfrankreich [1635–1661]: Ein Beitrag zur Verwaltungsgeschichte des Ancien Régime* [Münster: Aschendorff Verlag, 1980]); John Lynn, David Parrott, and Guy Rowlands (Review of *Giant of the Grand Siècle* and *The Wars of Louis XIV*, by John Lynn, *French History* 14 (2000): 450–54).

61. Sennewald, *Kursächsische Heer*, 595.

62. Perjés, “Army Provisioning,” 17–19.

63. For instance, SHStAdr 11237 10841/12.

the strain it put on the countryside and civilian workforce. The flexible and responsive tactics of the late Thirty Years' War, adapted to an environment of great distances and poor resources, were all easier for a smaller, more cavalry-heavy force to accomplish. From 1640 until 1650, the strength of the Saxon army oscillated around the manageable numbers of about 3,000 cavalry horses, 3,500 infantry, and 500 dragoons: somewhat fewer than 7,000 total.

At the end of the war Saxony's army was far smaller than the Imperial army (42,300) as of 1646, or the Bavarian army (20,563).<sup>64</sup> But it was much larger than the army of the neighboring territory of Brandenburg—hit far harder than Saxony had been by the war—which numbered 4,650 in 1640 and 1,800 in 1653.<sup>65</sup> Meanwhile, the army the duchy of Parma had raised fifteen years earlier had numbered about 7,000 on enlistment, but it could not maintain its strength, sinking sharply to about 1,000.<sup>66</sup> In terms of army strength, Saxony was a mid-level power like Savoy-Piedmont, whose army numbered about 6,000 in 1686.<sup>67</sup> The Saxon army was demobilized in 1650–51.

Any discussion which rests on claims about the size of seventeenth-century armies must first make sure these claims are as accurate as possible. Like work on other states, this article demonstrates for Saxony that it is possible to more accurately determine the strength even of seventeenth-century armies with enough low-level sources produced by armies themselves. Analyzing troop strength sheds new light on the operations of the Saxon army, revealing that events which were famous at the time or considered important by later historians, like the battle of Breitenfeld, may not have been as important as less-studied events like the battle of Wittstock or the Imperial/Saxon Pomeranian campaign. The Elector of Saxony's fiscal response to the war was not the modernization or centralization of his dominions but a series of expedients as Saxony lurched from crisis to crisis. But despite the destructive effects of the Thirty Years' War, Saxony not only survived, but did better than is commonly thought. The Electorate's continued relevance was bolstered by its ability to maintain an army in being and its willingness to use that army on behalf of its allies.

After 1635 Saxon forces took part in many battles on the side which most historians describe only as "Imperial," such as the catastrophic battle of Janckau in March 1645, which ruined at least two Saxon cavalry regiments.

Although the overall trend in Europe during the seventeenth century was toward large standing armies and more powerful states, Saxony's example shows this was far from a deterministic process. There was no single fiscal-military path to modernity. If we consider states like the Polish-Lithuanian Commonwealth, which collapsed, or Muscovy, which fell into civil war, this point is even clearer. Sweden, possibly the best example of a military revolution, became a centralized

64. Wilson, *Europe's Tragedy*, Table 6, 770; Kapser, *Bayerische Kriegsorganisation*, 220.

65. Jeremy Black, *European Warfare, 1660–1815* (London: Routledge, 1994), 101.

66. Hanlon, *The Hero of Italy*, 1.

67. Black, *European Warfare*, 102.

state organized for resource extraction and warfare under the Vasa dynasty. As of 1648, the total size of its army may have been as great as 90,000, while during the war of 1655–60, it totalled between 55,000 and 70,000 soldiers—although these numbers are probably high.<sup>68</sup> After 1635, the French army was larger at the conclusion of each new war than it had been in the last year of peace before it.<sup>69</sup> But in Central Europe many polities disbanded their armies as soon after peace as possible; during the 1650s and 1660s we see small numbers of garrison and guard troops, or repurposed militia. Saxony's mobilizations after the Thirty Years' War were small and took place within coalition endeavors. A force of 3,000 was raised in 1663 for the Imperial effort in the Austro-Turkish War, and in 1673 Saxony also sent a contingent to support the Emperor in the Franco-Dutch War. The Saxon standing army—permanent, regular regiments—dates only from 1682.<sup>70</sup>

Seventeenth-century Saxony did not permanently increase the size of its forces, nor was the electoral administration centralized or “modernized.” Instead, in this Electorate a pre-seventeenth-century means of raising money and troops bumped up hard against the limit of its carrying capacity. Saxony was capable of raising a field army of 36,000 but not of maintaining it for long, especially given the impacts of setbacks like Wittstock or disasters like the Pomeranian campaign. But it could maintain a smaller force, which it did until the end of the Thirty Years' War. Electoral Saxony's permanent standing army was established in 1683, but it was not until the reign of Augustus the Strong (1694–1733) that Saxony saw substantial centralization.

### Appendix

Strength numbers for all Saxon mercenary forces from 1618 to 1625 are listed in Table 1, cavalry strengths from 1631 to 1651 are in Table 2, infantry strengths from the same period are in Table 3, and dragoons from 1632 to 1651 are in Table 4. Each row in these tables depicts a single unit. Militia, garrison troops, feudal levies, the personal guards of important figures, and the two “old Saxon” regiments that Saxony paid for in the Imperial army after 1635 are not included.

The numerical values in these tables are the unit strengths for particular days. If a month for a particular unit contains asterisks rather than a number, that means that no data survive for that month for that unit, but the unit is still in being. To save horizontal space, months or years for which no data survive for any unit are not listed at all; for instance, Table 3 has only one month listed for 1643, because the only surviving troop count for infantry from that year was made in July. Because the tables are published online instead of on paper, the values are presented in full detail, tracking fluctuations in regimental strength day by day.

68. Jan Glete, *War and the State in Early Modern Europe: Spain, the Dutch Republic, and Sweden as Fiscal-Military States* (London: Routledge, 2001), 208.

69. Lynn, “The trace italienne and the Growth of Armies,” 299.

70. Reiner Gross, *Geschichte Sachsens* (Leipzig: Edition Leipzig, 2002), 1.

Seventeenth-century units were frequently named after their commanders (in the case of the Ungar Regiment of dragoons/cavalry, their captain's nickname). When a unit changed commanders, it changed names as well. These units are listed under the names they had at formation, which may not be the same as the names they had later.

These tables not only contain strength information, but also note some events like exceptionally damaging battles, a bad campaign, or an Oberst quitting to go into the service of the Elector of Brandenburg. These details are included to give the flavor of daily life in the running of a seventeenth-century unit or allow an attentive reader to track the slow roll-out of a troop surge, but are not necessary to the comprehension of my argument.

When the numbers in these tables are added up, we obtain the approximate total strength of the Saxon army throughout the Thirty Years' War, which is graphed in the article. Those values are the result of extrapolation. Although on several occasions the entire Saxon army or a branch of it counted its troops all at once (such as the cavalry before the battle of Wittstock in September 1636, or the entire army in March 1637), for much of the war data survive for only a few units each month. Fortunately, complete data sets exist for 1618–19 and 1631, which provide starting figures for Saxon troop strength in two different stages of the conflict. A running total of approximate troop counts after these years is obtained by adding the most recent surviving unit numbers, updating them each time a value changes.

Due to space limitations, the tables have been published on the Society for Military History website, at: <http://www.smh-hq.org/jmh/electoralsaxony.html>.

SMH members who do not have Internet access may contact the *JMH* office to request printed copies.

**Table 1:** Strengths of Saxon cavalry and infantry units from 1618 to 1625. Cavalry values are given in number of horses, infantry values in number of soldiers. Sources: SHStAdr 11237 10831/1 *Churf. Durchl. Zu Sachsen etc Erste und Andere Kriegsverfassung Nach entstandener Unruhe im Königreich Böhmen*; SHStAdr 11237 10831/2 *Kriegs Verfassung de Anno 1618 usq Anni 1650 vol II*; and muster rolls from the 1620s and late teens from SHStAdr 11237 series 10839, 10840, and 10841.

**Table 2:** Strengths of Saxon cavalry units from 1631 to 1651. Values given in number of horses. Units listed under their names at time of mustering, and in the order in which they appear in SHStAdr 11237 10831/1. Sources: SHStAdr 11237 10831/1 *Churf. Durchl. Zu Sachsen etc Erste und Andere Kriegsverfassung Nach entstandener Unruhe im Königreich Böhmen*; SHStAdr 11237 10831/2 *Kriegs Verfassung de Anno 1618 usq Anni 1650 vol II*; and Roland Sennewald, *Das Kursächsische Heer im Dreissigjährigen Krieg 1618–1648*, 549–80; muster rolls from the 1640s (several documents from SHStAdr 11237, series 10841).

**Table 3:** Strengths of Saxon infantry units from 1631 to 1651. Values given in number of soldiers. Units listed under their names at time of mustering, and in the order in which they appear in SHStADr 11237 10831/1. Sources: SHStADr 11237 10831/1 *Churf. Durchl. Zu Sachsen etc Erste und Andere Kriegsverfassung Nach entstandener Unruhe im Königreich Böhmen*; SHStADr 11237 10831/2 *Kriegs Verfassung de Anno 1618 usq Anni 1650 vol II*; and Roland Sennewald, *Das Kursächsische Heer im Dreissigjährigen Krieg 1618–1648*, 549–80; muster rolls from the 1640s (several documents from SHStADr 11237, series 10841).

**Table 4:** Strengths of Saxon dragoon units from 1632 to 1651. Values given in number of dragoons. Units listed under their names at time of mustering, and in the order in which they appear in SHStADr 11237 10831/1. Sources: SHStADr 11237 10831/1 *Churf. Durchl. Zu Sachsen etc Erste und Andere Kriegsverfassung Nach entstandener Unruhe im Königreich Böhmen*; SHStADr 11237 10831/2 *Kriegs Verfassung de Anno 1618 usq Anni 1650 vol II*; Roland Sennewald, *Das Kursächsische Heer im Dreissigjährigen Krieg*, 280; muster rolls from the 1640s (several documents from SHStADr 11237, series 10841); and strength tables from the 1640s (SHStADr 11237 10841/13).